SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME:Deep Learning COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour
Total Marks:50

Annexure A

Error Analysis Task

- A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

I ____logesh S 192525023________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided

Neural Network Performance Analysis Using Mean Squared Error (MSE)

Given

Training MSE = 0.85

Testing MSE = 5.20

Mean Squared Error (MSE) measures the average squared difference between the actual and predicted values. A lower MSE indicates better model performance.

The training MSE (0.85) is much lower than the testing MSE (5.20). This significant difference indicates that the neural network performs very well on the training data but poorly on unseen test data. This situation is commonly known as overfitting.

1. Understanding the Difference Between Training and Testing MSE
Training MSE
Training MSE is the error calculated using the same dataset that was used to train the neural network. A low training MSE means the model has learned the training data effectively.

Testing MSE
Testing MSE is the error calculated on new, unseen data. It measures how well the trained model generalizes to real-world data.

Interpretation
Since

Training MSE = 0.85

and

Testing MSE = 5.20

the testing error is much larger than the training error. This indicates that the model has learned the training data too closely, including its noise and unnecessary details, resulting in poor performance on new data.

2. Possible Reasons for the Difference
A. Overfitting
Overfitting is the most common reason for such a large difference between training and testing errors.

In overfitting:

The model memorizes the training data instead of learning general patterns.
It performs extremely well during training.
It fails to make accurate predictions on unseen data.
Characteristics

Low training error
High testing error
Poor generalization ability
B. Small Training Dataset
If the training dataset is too small:

The model does not see enough examples.
It cannot learn all possible patterns.
It memorizes the available data instead of learning general rules.
As a result, testing performance decreases.

C. Complex Neural Network Architecture
A network with:

Too many hidden layers
Large number of neurons
Excessive parameters
can easily memorize the training data.

Although training accuracy becomes very high, testing performance suffers.

D. Noise in the Training Data
Training data may contain:

Incorrect values
Missing values
Outliers
Measurement errors
The neural network may learn these noisy patterns, reducing its ability to predict new data correctly.

E. Lack of Regularization

Regularization techniques prevent the model from becoming overly complex.

If regularization is not used:

Weights become too large.
The model memorizes training data.
Overfitting occurs.
F. Poor Data Preprocessing
If data is not properly prepared:

Different feature scales
Missing values
Duplicate records
Incorrect encoding
the neural network may not learn efficiently.

G. Improper Train-Test Split
If training and testing datasets have different distributions:

Training data may be easier.
Testing data may contain unseen patterns.
This increases testing error.

H. Excessive Number of Training Epochs
Training for too many epochs causes:

Continuous reduction in training error.
Increase in testing error.
The model begins to memorize the training data.

I. Hyperparameter Selection
Incorrect hyperparameters can affect performance.

Examples include:

Learning rate
Batch size
Number of hidden layers
Activation functions
Optimizer choice
Poor hyperparameter selection may result in poor generalization.

3. Corrective Measures
A. Apply Regularization
Regularization reduces model complexity.

Common methods include:

L1 Regularization
L2 Regularization
Weight Decay

Benefits:

Prevents large weights.
Reduces overfitting.
Improves testing accuracy.
B. Use Dropout
Dropout randomly disables some neurons during training.

Advantages:

Prevents neuron dependency.
Improves generalization.
Reduces overfitting.
C. Early Stopping
Monitor validation error during training.

Stop training when validation error begins increasing.

Benefits:

Prevents over-training.
Saves computation time.
Produces a better model.
D. Increase Training Data
More data allows the model to learn a wider variety of patterns.

Methods include:

Collect additional data.
Data augmentation (where applicable).
Benefits:

Better generalization.
Reduced overfitting.
E. Simplify the Neural Network
Reduce:

Hidden layers
Number of neurons
Trainable parameters
A simpler model is less likely to memorize training data.

F. Improve Data Quality
Perform data preprocessing:

Remove duplicate records.
Handle missing values.
Remove outliers.
Normalize or standardize features.
Correct incorrect values.
High-quality data leads to better predictions.

G. Cross-Validation
Use techniques such as k-fold cross-validation.

Benefits:

Better estimation of model performance.
More reliable evaluation.
Improved hyperparameter tuning.
H. Hyperparameter Tuning
Optimize parameters such as:

Learning rate
Batch size
Number of epochs
Activation functions
Optimizer
Methods include:

Grid Search
Random Search
Bayesian Optimization
I. Feature Selection
Remove irrelevant or redundant input features.

Benefits:

Reduces complexity.
Faster training.
Better prediction accuracy.

4. Expected Outcome After Applying Corrective Measures
After implementing the above improvements:

Training and testing MSE become closer.
Generalization improves.

Prediction accuracy increases.
Overfitting is minimized.
The neural network performs better on unseen data.